CODE SPORT

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In this month's column, we continue our discussion on detecting duplicate questions in community question answering forums.

Based on our readers' requests to take up a real life ML/NLP problem with a sufficiently large data set, we had started on the problem of detecting duplicate questions in community question answering (CQA) forums using the Quora Question Pair Dataset.

Let's first define our task as follows: Given a pair of questions <Q1, Q2>, the task is to identify whether Q2 is a duplicate of Q1, in the sense that, will the informational needs expressed in Q1 satisfy the informational needs of Q2? In simpler terms, we can say that Q1 and Q2 are duplicates from a lay person's perspective if both of them are asking the same thing in different surface forms.

An alternative definition is to consider that Q1 and Q2 are duplicates if the answer to Q1 will also provide the answer to Q2. However, we will not consider the second definition since we are concerned only with analysing the informational needs expressed in the questions themselves and have no access to answer text. Therefore, let's define our task as a binary classification problem, where one of the two labels (duplicate or non-duplicate) needs to be predicted for each given question pair, with the restriction that only the question text is available for the task and not answer text.

As I pointed out in last month's column, a number of NLP problems are closely related to duplicate question detection. The general consensus is that duplicate question detection can be solved as a by-product by using these techniques themselves. Detecting semantic text similarity and recognising textual entailment are the closest in nature to that of duplicate question detection. However, given that the goal of each of these problems is distinct from that of duplicate question detection, they fail to solve the latter problem adequately. Let me illustrate this with a few example question pairs.

Example 1

Q1a: What are the ways of investing in the share market?

Q1b: What are the ways of investing in the share market in India?

One of the state-of-art tools available online for detecting semantic text similarity is SEMILAR (*http://www.semanticsimilarity.org/*). A freely available state-of-art tool for entailment recognition is Excitement Open Platform or EOP (*http://hlt-services4.fbk.eu/eop/index.php*). SEMILAR gave a semantic similarity score of 0.95 for the above pair whereas EOP reported it as textual entailment. However, these two questions have different information needs and hence they are not duplicates of each other.

Example 2

Q2a: In which year did McEnroe beat Becker, who went on to become the youngest winner of the Wimbledon finals?

Q2b: In which year did Becker beat McEnroe and go on to become the youngest winner in the finals at Wimbledon?

SEMILAR reported a similarity score of 0.972 and EOP marked this question pair as entailment, indicating that Q2b is entailed from Q2a. Again, these two questions are about entirely two different events, and hence are not duplicates. We hypothesise that humans are quick to see the difference by extracting the relations that are being sought for in the two questions. In Q2a, the relational event is "<McEnroe (subject), beat (predicate), Becker (object)> whereas in Q2b, the relational event is <Becker (subject), beat (predicate), McEnroe (object)> which is a different relation from that in Q2a. By quickly scanning for a relational match/mismatch at the cross-sentence level, humans quickly mark this as non-duplicate,